

# Contrastive Projection: Reading Transformer Internals by Differencing Logit Lenses

Olli Tuomi  
Evident Solutions Oy

## Abstract

Subtracting the hidden states of two matched inputs at each layer and projecting through the unembedding matrix  $W_U$  produces a training-free, layer-by-layer readout of what separates them in token space. The subtraction cancels the component the two inputs share and reads what differs: content that is buried below shared signals in the logit lens of either input alone.

We validate the method in two ways: injecting the difference vector at a single layer recovers the target’s prediction, while an equal-norm, same-frame near-neighbour contrast recovers nothing, and dose-response confirms directional specificity. For compositional contrasts, multi-contrast triangulation recovers token-readable causal content that single-pair readout misses (1%  $\rightarrow$  77% for “caught a cold”).

Applied to Phi-2 (2.7B), the method traces compound-noun recognition to a two-hop MLP $\rightarrow$ attention circuit, traces both the IOI and factual-recall circuits from their source token to the heads that carry the answer via position-resolved patching, reads what the model retrieves for real versus fictional entities, reveals that parametric knowledge triggers correction rather than override when contradicted, traces the divergence between bare and pragmatically marked quantifiers, and provides injection evidence that metaphor is per-domain routing rather than a figurativity flag, across 15 semantic axes and four models. We release code and data.

## 1 Introduction

A transformer processing “The hot dog was” predicts continuations about food. The same model given “The cold dog was” predicts continuations about an animal. The logit lens (nostalgebraist, 2020) projects each hidden state through  $W_U$  and at intermediate layers reads the same function words for both — “not, no, made, a, more.” The food distinction is not visible: it is present in the residual stream but outranked by other signals that dominate the projection.

Subtracting one hidden state from the other before projecting through  $W_U$  cancels the shared signals and reads what differs. At layers 8–24 for the hot-dog case, the contrastive projection reads “fried, crispy, delicious, flavor” — food vocabulary absent from either constituent’s logit-lens top-20 (0–1/5 overlap through L24). Two closely matched inputs share most of their computation; what survives the subtraction is the component where the model treats them differently.

Because it reads a *difference*, every contrastive reading is intrinsically pair-specific: it reports what separates one particular pair of prompts. A tight pair that varies a single thing may read a feature the model itself carries; a looser pair blends several differences at once, and  $W_U$  may not have a token for the blend. The reading is therefore steered by contrast design: what it surfaces is chosen, not discovered.

The arithmetic is the same as computing a RepE/ActAdd steering vector (Zou et al., 2023; Turner et al., 2023) and reading it through a logit lens (nostalgebraist, 2020). In this work the same arithmetic is applied systematically to *reading*, with two specific techniques built on it:

1. **Systematic trajectory reading.** Per-position tracing locates where a distinction first appears and how it flows between positions. Per-head decomposition identifies which attention heads carry it. Layer-by-layer readout tracks how the content changes from early detection to final prediction.
2. **Multi-contrast triangulation.** A single contrastive pair may produce uninterpretable tokens when the target signal is mixed with pair-specific content. Contrasting the same target against multiple baselines and averaging isolates the shared causal component. This recovers token-readable causal content from cases where single-pair readout fails: “caught a cold” recovers 1% from a single pair, 77% from five-baseline triangulation (reading “doctor, doctors, rest”).

We apply the method to Phi-2 (2.7B), with cross-model replication on Pythia-410M and Pythia-1.4B; the 15-axis taxonomy additionally covers Phi-4 (14B). The cases presented are selected from several hundred contrastive readings; we report the most illustrative.

## 2 Method

### 2.1 Contrastive projection

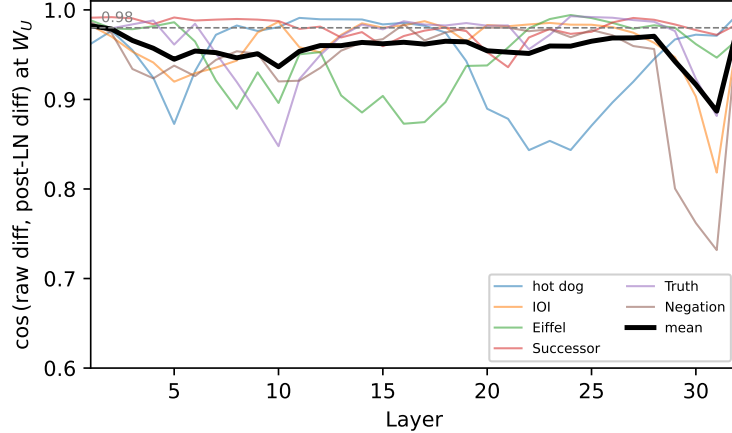
Given two inputs  $c$  and  $k$ :

1. Run both and extract hidden states at every layer at the read position:  $h_c[L]$  and  $h_k[L]$  for  $L = 0, \dots, N$ .
2. Compute  $\Delta h[L] = h_c[L] - h_k[L]$ .
3. Project:  $\Delta \text{logits}[L] = \Delta h[L] \cdot W_U^\top$ .
4. Read the top- $K$  most positive tokens (associated with input  $c$ ) and the top- $K$  most negative tokens (associated with input  $k$ ). The two poles together describe the contrast in token space.

No parameters are fitted. The choices are the input pair, the read position, and  $K$ .

**LayerNorm.** The model’s forward pass applies a final LayerNorm ( $\text{LN}_f$ ) before  $W_U$ . We project raw hidden states, bypassing  $\text{LN}_f$ . This is safe because we project a *difference* of two matched states: matched inputs have near-identical mean and standard deviation, so  $\text{LN}_f$ ’s centering and

scaling nearly cancel in the subtraction, and its learned shift cancels exactly. What remains is a per-dimension rescaling, which could in principle reorder the token ranking. Empirically it does not: across the six poster cases (IOI, hot dog, factual recall, successor, truth, negation), the cosine between the raw difference projection  $(h_c - h_k) \cdot W_U^\top$  and the strictly normalized variant  $(\text{LN}_f(h_c) - \text{LN}_f(h_k)) \cdot W_U^\top$  averages  $\approx 0.95$  per layer (mostly  $> 0.94$  per case), and the two produce the same top-5 tokens (Figure 1). The exceptions are occasional late-layer dips in individual cases (negation falls to 0.73 at L31), where the per-dimension rescaling reorders the ranking tail more.



**Figure 1:** Cosine between the raw ( $\text{LN}_f$ -bypassed) and strictly normalized difference projections at  $W_U$ , per layer, for the six poster cases (thin) and their mean (thick). L0 is omitted (the read-position token is shared, so the differenced state is zero).

Raw and normalized projections diverge in *magnitude*: residual-stream norms grow across layers (from 4 at L1 to 175 at L31), so raw  $\Delta \log$ its norms are not comparable across layers. Where we report norms across layers (e.g., §5.3), we use the relative norm  $\|\Delta h\| / \|h_c\|$  to remove this scale artifact. Bypassing  $\text{LN}_f$  means the projection relies on directional alignment between the residual stream and  $W_U$  rather than the calibrated magnitude that  $\text{LN}_f$  would provide; relative magnitude comparisons across layers should be interpreted accordingly.

**Why  $W_U$ , not  $W_E$ .** The input embedding matrix  $W_E$  is an alternative projection target — fc1 reads from the residual stream, which originates as  $W_E$  embeddings. We test both. In Phi-2, the row vectors of  $W_E$  and  $W_U$  for identical tokens are nearly orthogonal (mean cos 0.005 across the vocabulary). Projecting known clean MLP neurons’ fc1 rows through  $W_U$  produces interpretable labels (N925: “food, food, foods”; N7828: “food, edible, delicious”; N841: “husband, husbands, boyfriend”). The same rows through  $W_E$  produce noise (“union, replacement, abet”; “required, when, executive”; “Representative, postal, learning”). Contrastive  $\Delta h$  shows the same pattern:  $W_U$  reads “tast, charred, crispy, delicious” for hot dog at L28;  $W_E$  reads “is, ern, erton.” The alignment gap widens with depth: fc1 rows’ peak token-specificity via  $W_U$  exceeds  $W_E$  by  $1.02\times$  at L4, growing to  $1.40\times$  at L28.

## 2.2 Per-position reading

By reading the contrast at every position, we trace information flow: at which position does a meaning distinction first appear? Does it appear at the differing token itself, or at a later position that attends to it?

## 2.3 Attention and MLP decomposition

Each transformer layer adds two components to the residual stream: attention output and MLP output. We capture both via forward hooks and compute their contrastive projections separately. This identifies which component writes a given content distinction at each layer — without causal intervention.

We note that a large contrastive norm in either component indicates that the component’s output differs between the two inputs, not that the component causes the distinction. Causal verification (activation patching) is required to establish causality.

## 2.4 What the projection reads

The contrastive projection re-ranks the logit lens through the input design rather than through a learned decomposition. A raw logit-lens readout of either input is dominated by signals the two inputs share (often high-frequency function words at the top of the ranking); the subtraction cancels the shared component and reads the remainder.

$W_U$  projection assigns token labels to directions in the residual stream. At intermediate layers, these labels name the axis of variation the contrast isolates: they localize where a distinction becomes legible in token space. They do not certify its cause, and they are not an account of everything the state carries; causal claims follow only from the injection and patching tests below. The token labels are not predictions of what the model will output at that layer.

**Single-pair vs. multi-contrast readout.** A single contrastive pair isolates one axis of variation, but the readout mixes the target signal with content specific to that baseline. When the target concept is a single entity (a name, a city), the pair-specific content is small and the readout is clean. When the target is compositional (a food compound, a moral judgment, a quantifier’s pragmatic force), pair-specific content can dominate the readout, producing uninterpretable tokens. Each baseline contributes its own legitimate signals — “fish” adds fishing-related content, “bus” adds transportation content — alongside the shared illness signal. Multi-contrast triangulation — contrasting the same target against several baselines and averaging — cancels the signals that vary across baselines and preserves the shared causal component.

We verify this empirically in §3.3.

**Read position and stability.** Contrastive reads taken at the very start of a prompt are unstable: semantically irrelevant changes, such as capitalising the first word, can shift the readout. A shared

preamble, identical across the pair and therefore cancelled by the subtraction, moves the read into a stable region where the target contrast dominates. We do not determine why the start is unstable: it may reflect the attention-sink and massive-activation structure present near the beginning of a sequence (Sun et al., 2026), or the model still settling register and topic before the small target difference can dominate. The examples in this paper use bare prompts for clarity. For a well-separated lexical contrast the preamble is only a refinement: the distinction survives without it (food vocabulary dominates the late-layer readout either way) and merely the specific tokens shift. It can, however, be essential: read at a position where the two continuations are about to diverge, a bare contrast can decode to register and next-token soup, while reading the same pair inside a shared frame, a few tokens away from that divergence, recovers a legible causal reading.

### 3 Validation

#### 3.1 The full activation delta is causal

The probe is exploratory. As a baseline, we test whether the full activation delta  $\Delta h[L] = h_{\text{context}}[L] - h_{\text{control}}[L]$  contains the causal information by injecting it into the control’s residual stream at layer  $L$  and measuring prediction gap recovery. This is standard activation patching: adding  $\Delta h$  to the control effectively replaces its activation with the context’s at that layer, so recovery at late layers is expected. The interesting result is the *layer profile* — at which layer does recovery onset, and does this match the contrastive readout?

| Case                            | L4 | L12 | L20 | L24 | L28  | L31  | Peak $z$ |
|---------------------------------|----|-----|-----|-----|------|------|----------|
| hot dog $\rightarrow$ delicious | 0% | 5%  | 73% | 80% | 133% | 150% | 4249     |
| IOI $\rightarrow$ Mary          | 0% | 0%  | 0%  | 5%  | 75%  | 113% | 525      |
| Eiffel $\rightarrow$ Paris      | 0% | 0%  | 2%  | 55% | 73%  | 66%  | 1778     |
| Successor $\rightarrow$ Tuesday | 0% | 0%  | 0%  | 11% | 87%  | 98%  | 223      |

**Table 1:** Prediction-gap recovery from injecting the full activation delta  $\Delta h[L]$  at one layer, for four cases (percent of the clean-control gap). Random directions of equal norm give zero mean recovery; the final column reports the  $z$ -score against that null.

Recovery is zero at early layers and clear by L28–31 (75–150%). Recovery above 100% means that the injected  $\Delta h$  overshoots the target prediction. This is expected:  $\Delta h$  captures the full difference between the two residual streams, but the control’s own residual stream at late layers already encodes partial prediction content, so adding the full difference on top of that partial encoding exceeds the target. Random directions of the same norm give zero mean recovery at every layer ( $z = 223$ – $4249$ ). This establishes that the *direction* of  $\Delta h$  matters, not just its norm — but a random direction is a weak null: it is near-orthogonal to every meaningful direction, so beating it is easy. The near-neighbour control below is the sharper test, and Section 3.3 tests whether the token-readable component of  $\Delta h$  carries the causal content.

**Near-neighbour contrasts.** A sharper test replaces the random direction with a *same-frame near-neighbour* contrast of equal norm. For the hot-dog case, we inject  $\Delta h$  from “The X dog was” minus

“The cold dog was,” norm-matched to the true contrast, for seven adjectives. Non-food adjectives (warm, big, wet, angry) recover  $\approx 0\%$  at every layer. “Delicious” also recovers  $\approx 0\%$  despite being a food word: the compound recognition is specific to “hot dog,” not to any food adjective before “dog.” “Fried” partially recovers (12–20%,  $\cos = 0.81$  with the true direction at L28), consistent with “fried dog” sharing part of the food-compound path. Recovery is specific to the content of the contrast, not a generic effect of adding a same-norm structured direction at that position.

**Recovery onset.** Hot dog recovery begins at L12 (5%), rising sharply at L20 (73%) — the layers where food vocabulary appears in the contrastive projection at the prediction position. IOI recovers from L24 (where “Mary” first dominates the readout). Factual recall recovers from L20 (where “France” first appears).

### 3.2 Dose-response and bidirectional causality

The layer-sweep above injects the full  $\Delta h$  at one layer. We can also extract a specific contrastive *direction* from multiple pairs and test its causal effect with controlled magnitude.

**Eatability direction.** We extract a food-compound direction by averaging the contrastive (hot dog minus  $X$  dog) for  $X \in \{\text{cold, angry, old, pet, stray}\}$  at the dog position, L4 post. These five directions are mutually consistent (pairwise  $\cos$  0.72–0.84). We inject the mean direction into non-food prompts at the dog position at varying fractions of its natural magnitude. Note that cold dog and angry dog are in the extraction set; the injection test measures whether the averaged direction produces graded, dose-dependent effects on model output, not whether it generalizes to held-out prompts. The multi-contrast triangulation experiments (§3.3, Table 6) provide the held-out generalization test, where the shared component is extracted from one set of baselines and recovery is measured on a different baseline.

Greedy generation shows the induced shift is semantic, not a single-token artifact:

| Prompt            | Injection | Greedy continuation                                                                               |
|-------------------|-----------|---------------------------------------------------------------------------------------------------|
| The cold dog was  | baseline  | shivering in the snow, so the owner put a sweater on it.                                          |
| The cold dog was  | +0.50×    | more popular than the hot dog because the cold dog was easier to eat.                             |
| The cold dog was  | +1.00×    | more expensive than the hamburger because the hamburger was made with better quality ingredients. |
| The angry dog was | baseline  | barking loudly at the mailman.                                                                    |
| The angry dog was | +1.00×    | more expensive than the hamburger because the hamburger was cheaper.                              |

**Table 2:** Greedy continuations under eatability injection, confirming a coherent semantic shift rather than a single-token artifact.

Subtracting the same direction from “The hot dog was” reverses the effect (Table 3):

At +1.0×, “The cold dog was” generates a food-comparison sentence. At −1.5×, “The hot dog was” generates an animal-care sentence. The minimum dose that flips top-1 prediction is 0.30×

| Prompt          | Injection     | Greedy continuation                                                     |
|-----------------|---------------|-------------------------------------------------------------------------|
| The hot dog was | baseline      | too spicy for the child...                                              |
| The hot dog was | $-1.00\times$ | more popular than the cold dog because the hot dog was more appetizing. |
| The hot dog was | $-1.50\times$ | panting heavily, so the owner gave it some water to cool down.          |

**Table 3:** Subtracting the eatability direction from “The hot dog was” reverses the effect (greedy continuations).

for cold dog (where “cold” partially aligns with food) and  $0.65\times$  for angry dog (where “angry” strongly primes the animal reading).

**Truth direction.** We extract a truth/falsity direction by averaging the contrastive (true statement minus false statement) for five fact pairs (Paris/France, water/ $100^\circ$ , Sun/star, dogs/mammals, Tokyo/Japan) at the prediction position. As with the eatability demonstration, the statements injected below overlap the extraction set; the leave-one-out replication at the end of this section provides the held-out test. Injecting into false statements (Table 4):

| Prompt                                                | Injection           | Greedy continuation                    |
|-------------------------------------------------------|---------------------|----------------------------------------|
| Paris is not the capital of France. This statement is | baseline            | false. Paris is the capital of France. |
| Paris is not the capital of France. This statement is | $+0.50\times$ truth | true.                                  |
| Paris is not the capital of France. This statement is | $+1.00\times$ truth | true.                                  |
| Dogs are reptiles. This statement is                  | baseline            | false.                                 |
| Dogs are reptiles. This statement is                  | $+1.00\times$ truth | true.                                  |

**Table 4:** Injecting the averaged truth direction into false statements flips the model’s verdict (greedy continuations).

Subtracting from true statements (Table 5):

| Prompt                                            | Injection          | Greedy continuation                                                         |
|---------------------------------------------------|--------------------|-----------------------------------------------------------------------------|
| Paris is the capital of France. This statement is | baseline           | true.                                                                       |
| Paris is the capital of France. This statement is | $-1.0\times$ truth | false. Paris is the capital of France, but it is not the only city...       |
| The Sun is a star. This statement is              | baseline           | true because we know that the Sun is a star based on scientific evidence... |
| The Sun is a star. This statement is              | $-1.0\times$ truth | false. The Sun is a star, but it is not the only star in the universe...    |

**Table 5:** Subtracting the truth direction from true statements flips the verdict (greedy continuations).

Scaling the extraction to 15 fact pairs and testing by leave-one-out (the direction is averaged from the other 14 pairs and injected at the final position of the held-out statement, extracted at L20) confirms the effect on held-out statements: at twice the natural magnitude, all 15 false statements flip to “true” and all 15 true statements to “false” (10/15 and 7/15 at natural magnitude).

The truth direction has lower cross-pair consistency (mean pairwise cos 0.36 across the 15 pairs) than the eatability direction (0.72–0.84), reflecting the fact that “what makes Paris-is-the-capital true” and “what makes dogs-are-mammals true” share less structure than different food-compound

contrasts. The lower consistency, and the need for above-natural magnitude to flip every held-out statement, suggest the truth direction is noisier than the lexical axes and may benefit from the multi-contrast triangulation of §3.3.

### 3.3 Multi-contrast triangulation recovers compositional signals

For compositional contrasts, single-pair readout may fail because the target signal is mixed with pair-specific content. We test whether multi-contrast triangulation recovers the causal component.

For “She caught a cold” vs “She caught a fish,” we project the single-pair  $\Delta h$  onto the subspace spanned by the  $W_U$  rows of its top-20 and bottom-20 tokens (40 directions total, orthogonalized via QR decomposition) and inject only this token-subspace component. This recovers 1% of the prediction gap — the token-readable part of the single-pair  $\Delta h$  is not where the causal content lives. (By contrast, injecting the full unrestricted  $\Delta h$  at the same layer recovers 69% — the causal content is present but not aligned with the loudest  $W_U$  directions.) The same case triangulated against five baselines yields an averaged  $\Delta h$  whose token-subspace component — constructed the same way, from the  $W_U$  rows of the averaged vector’s own top-20 and bottom-20 tokens — reads “doctor, doctors, rest, see” and recovers 77%. The comparison is therefore apples-to-apples: both numbers inject only the token-readable component. The five baselines are:

- “She caught a **fish** and went to” (physical catch)
- “She caught a **ball** and went to” (physical catch)
- “She caught a **bus** and went to” (transportation)
- “She caught a **thief** and went to” (apprehension)
- “She caught a **glimpse** and went to” (perception)

Each baseline shares the “caught a —” frame but differs in what was caught. The shared component across all five contrasts — what “cold” has that fish, ball, bus, thief, and glimpse do not — is the illness signal. Multi-contrast averaging isolates this from the pair-specific content (e.g., medical vs fishing in the cold/fish pair alone) that dominates the single-pair readout (Table 6).

| Case                   | Single-pair recovery | Multi-contrast recovery | Shared component reads  |
|------------------------|----------------------|-------------------------|-------------------------|
| IOI (Mary/John)        | 84%                  | — (not needed)          | Mary, mary              |
| Capital (Paris/Berlin) | 97%                  | — (not needed)          | Paris, French           |
| Caught a cold          | 1%                   | 77%                     | doctor, doctors, rest   |
| Theft/moral            | 4%                   | 75%                     | evade, hoped, evasion   |
| Hot dog food           | 11%                  | 49%                     | crispy, charred, cooked |
| Some/all (partitive)   | −40%                 | 101%                    | some, Some, SOME        |

**Table 6:** Single-pair  $W_U$  projection vs. multi-contrast triangulation. Entity-identity contrasts (top rows) read cleanly from a single pair. Compositional contrasts (bottom rows) require multiple baselines to isolate the token-shaped causal signal from pair-specific content.

**Why averaging works.** A companion paper derives the detection arithmetic for this readout (Tuomi, 2026): a token-aligned component carrying an energy fraction  $f$  of the difference vector surfaces in the top- $k$  only above a visibility threshold  $f^* \approx 2 \ln(2V)/d$  (about 0.9% of the vector’s energy for Phi-2), and averaging  $N$  matched difference vectors suppresses the incoherent remainder by  $\sqrt{N}$ , lowering the detection bar roughly  $N$ -fold. The single-pair failures in Table 6 are consistent with sub-threshold aligned energy rather than absent content; averaging does not sharpen the signal, it suppresses the unaligned remainder.

**Granularity.** Multi-contrast averaging recovers what is shared across baselines, but cannot determine whether the result is a single feature or a stable bundle of co-occurring features. Entity-identity contrasts (Mary/John) recover a component that corresponds to one token. Compositional contrasts (illness, food-compound) may recover a bundle: “doctor, rest, see” could reflect co-activated medical, recovery, and action features rather than a single “illness” feature. We do not claim to recover atomic features. The method recovers the shared causal component at the granularity the contrast design provides.

### 3.4 Null model: smoothness is structural, not semantic

$W_U$  contributes the interpretable token labels. Trajectory smoothness (consecutive-layer cosine  $\sim 0.9$ ) is a general property of the residual stream, not specific to meaningful contrasts: unrelated pairs (“The hot dog was” vs “Quantum mechanics is”) are equally smooth ( $0.93 \pm 0.01$ ,  $N = 40$ ) as minimal pairs ( $0.88 \pm 0.03$ ,  $N = 6$ ), because any two residual streams diverge gradually. Random directions give cosine  $\sim 0.00$  ( $z = 64\text{--}93$ ), confirming  $\Delta h$  is structured rather than noise, but smoothness alone does not validate content.

### 3.5 Baseline subtraction

A contrastive projection requires a second prompt. A generic alternative is to subtract the mean residual of many random text snippets of the same length (100 in our case), removing what is common to arbitrary English text at this position and leaving whatever is specific to the target. This achieves for a single prompt what the paired subtraction achieves for a pair: the raw logit lens is dominated by high-frequency function words (“a, the”) at every layer, burying content tokens below them; after baseline subtraction, the function words cancel and content surfaces.

Table 7 applies this to the hot-dog example read inside a shared preamble (§2.4). In the raw logit lens, food tokens rank 300–1200 at intermediate layers, invisible below function words. After baseline subtraction they surface to the top.

## 4 Lexical disambiguation

### 4.1 Compound noun: hot dog

Prompts:

| L  | Raw logit lens      | Baseline-subtracted                                      |
|----|---------------------|----------------------------------------------------------|
| 6  | not, only, always   | eaten <sup>+2</sup> , foods, menu                        |
| 10 | not, indeed, always | flavor <sup>+3</sup> , sold, cheese                      |
| 16 | not, only, always   | cheese <sup>+4</sup>                                     |
| 20 | not, more, only     | delicious <sup>+8</sup> , cheese, flavors                |
| 24 | not, more, a        | cooked <sup>+19</sup> , grilled, delicious               |
| 28 | not, a, the         | cooked <sup>+30</sup> , grilled <sup>+29</sup> , sausage |

**Table 7:** Raw logit-lens top tokens vs. baseline-subtracted top tokens (Phi-2, final position). The raw projection reads “not, only, always” at every layer (function words dominate). After subtracting the mean of 100 random text snippets, food vocabulary surfaces from L6 onward and builds through L28 — content that ranks 300–1200 in the raw projection.

- “The hot dog was” (food item)
- “The cold dog was” (cold animal)

**Predictions:** hot dog → “too” (0.088), “more” (0.080), “cooked” (0.063); cold dog → “sh[ivering]” (0.447), “shaking” (0.036).

**Per-position trace (4 tokens: The, hot/cold, dog, was):**

At position 2 (“dog”) — same token in both prompts, L0 contrast is zero:

- L1: “hot, molten, fiery” — “dog” receives temperature info from “hot”
- L4 (post-MLP): “fried” first appears — compound meaning recognized by MLP
- L5 (post-attention): “fried” persists — L5 attention adds little locally
- L24: “fried, seasoning, Flav, Serv, vendor” — full food vocabulary
- L32: “vendor, vendors, stand, topping” — hot dog stand

Sub-layer decomposition at L4–L5 confirms the compound is recognized by the MLP at L4, not by attention. The MLP contrastive norm at L4 (20.0) dominates the attention norm (3.8); “fried” appears in the MLP output but not the attention output. L5 attention writes orthogonally to the food direction ( $\cos(\Delta_{\text{pre}}, \Delta_{\text{attn}}) < 0.08$  across all contrasts tested).

At position 3 (“was”) — reads from “dog”:

- L5 (attention): “delicious, substitutes, cooked” — food signal arrives via attention from “dog” (one layer after MLP recognizes the compound at “dog”)
- L6: “fried, delicious, tasty, breakfast”
- L28: “tasty, charred, crispy, delicious, spicy” (hot) vs “grooming, shudder, shaking” (cold)

**Attention routing at L5, position “was”:** Head 19 attends to “dog” position with weight 0.911 in the hot-dog context vs 0.735 in cold-dog. Head 29 attends 0.345 vs 0.138. These heads read the compound-noun information from “dog” and write it to “was.”

**Multi-contrast convergence:** The food-compound direction is stable across reference points. Five different contrasts (hot dog minus cold / angry / old / pet / stray dog) produce pairwise cosine 0.72–0.84 at the dog position, confirming the signal is about food-compound identity, not temperature or emotion. Note that high pairwise cosine between full  $\Delta h$  vectors does not imply high recovery from their top  $W_U$  token directions (Table 6 reports 11% for hot dog). The cosine measures alignment of the full d-model vectors; the 11% measures how much of each vector lives in the subspace of its loudest  $W_U$  tokens. A signal can be consistent in direction (high cosine) while being distributed across many  $W_U$  directions rather than concentrated in a few (low token-subspace recovery).

**Neuron and head detail:** The MLP write at “dog” is distributed: the ten neurons pushing hardest along the food direction carry only  $\sim 10\%$  of it, spread across hundreds, some writing food (one writes “cooked, foods, dishes”) and some suppressing animal content (another writes “dogs, puppies”). The routing head L5.H19, already flagged above by its attention to “dog,” writes “cooked, eaten” into “was”: the edibility signal the read surfaces at the prediction position. The readable food content is thus genuinely computed, but by a distributed MLP write plus a localized routing head, not by any single neuron.

**Activation patching at multiple layers and positions** (Table 8):

| Position patched   | Layer  | P(food)   | Top-1    | Effect                       |
|--------------------|--------|-----------|----------|------------------------------|
| (baseline hot dog) | —      | 0.142     | “too”    | —                            |
| dog (pos 2)        | L0–L4  | 0.000     | “pant”   | Food eliminated              |
| dog (pos 2)        | L5     | 0.094     | “more”   | Food partially survives      |
| dog (pos 2)        | L12+   | 0.12–0.14 | “too”    | Near baseline                |
| hot (pos 1)        | L0–L28 | 0.11–0.14 | “too”    | Minimal effect at all layers |
| was (pos 3)        | L0–L5  | 0.13–0.14 | “more”   | Food preserved               |
| was (pos 3)        | L12    | 0.058     | “placed” | Food declining               |
| was (pos 3)        | L20+   | 0.005     | “sh”     | Food eliminated              |

**Table 8:** Activation patching of the hot-dog prompt at three positions and several layers, tracing the two-hop MLP→attention chain (P(food) and top-1 token).

The patching traces the two-hop chain precisely. (1) Patching “dog” at L0–L4 eliminates the food signal entirely — the compound meaning has not yet been computed. At L5, just after the L4 MLP write where “fried” first appears in the contrastive projection, food partially survives the patch — the computation is completing at these layers. By L12+ the patch has minimal effect. (2) Patching “hot” has no effect at any layer — its information has already been copied to “dog” via L0 attention. (3) Patching “was” has no effect at L0–L5 (the food information hasn’t arrived yet) but eliminates food from L12 onward — consistent with attention at L5–6 copying the compound meaning from “dog” to “was.”

**Mechanism:** Attention at L0 copies “hot” to the “dog” position. The MLP at L4 recognizes the compound at the “dog” position (“fried” first appears in the MLP output). Attention at L5 broadcasts this from “dog” to “was” (H19, attn=0.91). The contrastive projection identified each of these stages; activation patching at three positions and multiple layers confirms the information flow.

## 4.2 Other disambiguation cases

The same method traces disambiguation in verb-object pairs and noun ambiguity:

**“He caught a cold and” vs “He caught a fish and”** (verb meaning changes):

- L12: “contracted, virus, contagious” (cold pole)
- L28: “fever, coughing, cough, flu” vs “proudly, reel, bait, trout”

**“The bank was steep and” vs “The bank was closed and”** (noun disambiguated by adjective):

- L12: “climb, inclined, steep” (terrain pole)
- L27–28: “bank, banks, banking” reappear in the contrastive projection — the noun’s representation is revisited after the modifier has disambiguated it. This is compatible with a two-pass pattern (resolve meaning from the modifier, then update the noun representation), though the contrastive projection reads content, not mechanism.

**“He was fired up and” vs “He was fired and”** (particle changes meaning):

- L28: “excited, ready, energetic” (fired up) vs “sued, blacklist, lawsuits” (fired)

## 4.3 Scalar implicature: pragmatic inference in the residual stream

Lexical disambiguation (§4.1–4.2) traces how the model resolves word meaning. A different kind of distinction involves quantifiers: “Some of the students passed” and “All of the students passed” produce different continuations. We trace how the residual stream diverges between these two quantifiers.

**Design:** Contrast “Some of the students passed the exam, so” against “All of the students passed the exam, so” and read the contrastive projection at the final token.

**Trajectory (some pole +, all pole –)** (Table 9):

| Layer | Some pole reads         | All pole reads                        |
|-------|-------------------------|---------------------------------------|
| L8    | might, may              | except, together                      |
| L20   | Others, another         | everyone, except, Everyone            |
| L28   | others, Others, another | everyone, Everyone, everybody, except |

**Table 9:** Contrastive trajectory for the some/all quantifier contrast (positive pole = some, negative pole = all).

The negative pole consistently reads “everyone, everybody, except” from L8 onward — tokens associated with “all” contexts. This reflects the divergent prediction trajectories of the two quantifiers: the “all” context primes totality tokens, and the contrastive projection surfaces this divergence. Whether this reflects an internal “not everyone” computation or simply the output distribution difference between the two prompts cannot be determined from the contrastive readout alone.

**Cross-content consistency.** The same some/all contrast across 6 noun phrases (students, cookies, houses, employees, books, countries) produces pairwise cosine 0.12–0.45 (mean 0.30). The negative pole consistently reads totality words (“everyone, all, except, none”) across all noun phrases, while the positive pole varies with content. Using multi-contrast triangulation across noun phrases, the shared component reads “Others, Other, another” vs “all, everyone, none, except, everybody” — both poles token-shaped.

**Scalar gradient.** Projecting five quantifiers (None, Few, Some, Most, All) onto the some↔all axis at L28 (Table 10):

| Quantifier | Projection   |
|------------|--------------|
| None       | +27          |
| Few        | +14          |
| Some       | +84          |
| Most       | +13          |
| All        | 0 (baseline) |

**Table 10:** Projection of five quantifiers onto the some↔all axis at L28. “Some” is the outlier, consistent with its status as the pragmatically marked quantifier.

The gradient is not monotonic: “Some” is the outlier, projecting far beyond any other quantifier. “Few” and “Most” project similarly despite being semantically opposite. This is consistent with “some” being the pragmatically marked quantifier — the one that generates scalar implicature — rather than the scale reflecting quantity.

**Explicit vs bare.** Contrasting “Some of the students passed” against “Some but not all of the students passed” produces a non-trivial  $\Delta h$  (norm 57–59 at L28). The –side reads “some, Some, not, neither, but, BUT” — the explicit restriction changes the representation. The bare “some” computes partial implicature (the “everyone/except” signal), but adding “but not all” adds further restriction beyond what the bare form computes.

## 5 Replicating landmark findings

### 5.1 Indirect object identification

The IOI circuit (Wang et al., 2023) identifies how GPT-2 resolves which name a pronoun refers to. We replicate the core finding with the contrastive method.

**Design:** Contrast “John and Mary went to the store. John gave a book to” vs the same with names swapped. The IO name should appear in the contrastive projection at the prediction site.

**Prompts (example):**

- A: “John and Mary went to the store. John gave a book to” → Mary
- B: “Mary and John went to the store. Mary gave a book to” → John

**Contrastive trajectory:**

- L8: gender emerges (“himself, his” vs “herself, her”)
- L24: IO names appear (“Mary, Mary” vs “John, John”)
- L28: fully stable (“Mary, Mary, mary” vs “John, John, john”)

**Per-head decomposition at L24:** Decomposing the attention output by head identifies which heads carry the IO name. Three heads dominate consistently across three name pairs (John/Mary, Alice/Bob, Dan/Eve; Table 11):

| Head | John/Mary (norm) | Alice/Bob | Dan/Eve | Content                       |
|------|------------------|-----------|---------|-------------------------------|
| H14  | 8.3              | —         | 11.4    | IO name (MD/Mary, de/draw)    |
| H1   | 7.5              | 5.9       | 7.0     | IO name (Mary, Bob, ever/ves) |
| H16  | 3.4              | 4.5       | 12.8    | IO name (Mary, Bob, Eve)      |

**Table 11:** Per-head contrastive norm at L24 for the IOI name-swap contrast across three name pairs. Three heads carry the indirect-object name.

These heads have the largest contrastive norms at L24 and read IO-name tokens when projected through  $W_U$ . No other head exceeds norm 3.0 consistently. This identifies heads that carry the IO name signal — the same observational signature as the name-mover heads that Wang et al. (2023) found in GPT-2, here located in Phi-2 using only contrastive projection.

**Why the name-swap contrast cannot test routing.** Prompts A and B run the identical IOI algorithm — detect the duplicated subject, suppress it, copy the other name — and differ only in which tokens fill the roles. The shared algorithm cancels under subtraction, so the heads the contrast surfaces are name-content carriers, not the routing mechanism. Three tests sharpen what they are and whether they are causal.

**Attention patterns confirm two genuine name-movers.** At L24, H14 and H16 attend from the final position to the IO token (“Mary”), with 0.86 and 0.56 of their attention mass on it — reading the indirect object directly, the name-mover signature. H1, also flagged by contrastive norm, instead attends to the sequence-initial sink token (0.87): it is a false positive of the norm criterion that the attention pattern filters out. Pairing contrastive norm with attention direction separates the readers (H14, H16) from sink heads.

**Ablation and denoising agree: the heads are not the causal carrier.** Ablating all three heads at L24 (zeroing their pre-projection output at the last position) reduces  $P(\text{Mary})$  from 0.696 to 0.685 — a 1.1% drop; individual ablation gives H14 −0.2%, H16 −0.9%, H1 none. Three control heads (H0, H5, H10) give a comparable 0.3% drop and the generation is unchanged. Denoising is the more sensitive complement: we corrupt the prompt by swapping the giver ( $P(\text{Mary}) = 0.011$ ) and patch clean head outputs back at the final position, measuring how much of the clean-corrupt gap (toward 0.696) is restored. Patching the name-movers restores  $\leq 1\%$ ; patching all 32 heads at the final position restores  $\leq 2\%$  — at every layer from L20 to L28. Injection gives the same verdict for sufficiency: the name-movers’ contrastive output ( $A - B$ ) shifts  $P(\text{Mary})$  in B from 0.026 to only 0.037

at  $1\times$  and 0.067 at  $3\times$ , and patching all 32 heads  $A\rightarrow B$  reaches only 0.041. By contrast, injecting the full residual-stream  $\Delta h$  at L28 drives  $P(\text{Mary})$  to 0.642. The decision is carried by the cumulative residual stream, not by any layer’s attention write at the final position.

**A structure-preserving contrast surfaces a name-invariant signal.** Contrasting the IOI prompt against a no-duplicate baseline (giver = a third name, “Sam”) keeps the duplicate-detection computation in the subtraction. This surfaces different heads — at L26 one reads the giver name (“Sam”), others read structural tokens — and the resulting delta does not project to clean tokens through  $W_U$ .

$W_U$ -illegibility alone is weak evidence for structure, however: a token-aligned component can be present yet sit below the readout’s detection threshold (Tuomi, 2026). We test the claim directly with prompt design rather than a trained probe. Build a family of eight name-sets that share the IOI structure but vary the token identities (John/Mary, Alice/Bob, ...), and form both contrasts at END for each set. As depth increases the two diverge sharply: the name-swap (token) direction rotates with the names and collapses toward orthogonality across the family (mean pairwise cosine  $+0.73$  at L20  $\rightarrow +0.21$  at L24  $\rightarrow +0.01$  at L28), whereas the no-duplicate direction stays aligned ( $+0.83 \rightarrow +0.66 \rightarrow +0.37$ ). Averaging each contrast over the family — which cancels token-specific components and keeps the shared one — preserves 0.83 of the no-duplicate norm at L24 against 0.56 for the name-swap. The averaged name-swap survivor reads as  $W_U$  junk (its per-set reads are clean but different names — “Mary”, “Sam” — that cancel), while the averaged no-duplicate survivor reads as identity-free recipient-role content (“charity, share, shop, others”). Position-resolved attention agrees: heads H12 and H20 read  $\text{END} \rightarrow \text{duplicate-position}$  across all eight name-sets (0.16–0.37), keying on the *position* regardless of which name fills it — a structural signature that never touches  $W_U$ .

The signal is therefore present, contrast-isolable, and name-invariant where the name-identity content is not. (The surviving direction reads as the recipient role, which is broader than “inhibition” specifically; separating routing from recipient-role content would need a further control.)

**Tracing the path by position-resolved patching.** The corrupted (swapped) prompt differs from the clean one in a *single* input token — the second subject at position 9 (S2) — so all causal divergence originates there. Patching the full residual stream at one position from clean into the corrupted run, across layers, traces where the information lives (recovery of  $P(\text{Mary})$ , corrupt 0.011  $\rightarrow$  clean 0.696):

| Patched position | L0   | L8  | L12 | L16 | L20 | L24 | L28  |
|------------------|------|-----|-----|-----|-----|-----|------|
| IO (pos 3)       | 0%   | 0%  | 0%  | 1%  | 0%  | 0%  | 0%   |
| S2 (pos 9)       | 100% | 90% | 79% | 59% | 16% | 6%  | 0%   |
| END              | 0%   | 0%  | 0%  | 1%  | 37% | 59% | 98%  |
| S2 + END         | 101% | 95% | 88% | 87% | 96% | 97% | 100% |

**Table 12:** Position-resolved patching of the IOI prompt (recovery of  $P(\text{Mary})$ , corrupt  $\rightarrow$  clean). The causal signal transfers from S2 to END between L16 and L28.

The IO position never matters — its token is identical in both prompts. The S2 position carries all

of the recovery at L0 and loses it monotonically, while END gains it in lockstep; S2 and END jointly recover  $\sim 100\%$  at every layer. The causal signal transfers from S2 to END between L16 and L28, with the crossover (END-only overtaking S2-only) at L18–20.

Attention identifies the heads that perform the transfer. At L18–20, heads attending from END to S2 dominate (H10, H12, H20, H31) — they move the duplicate-subject signal forward. At L22–26, heads attending to the IO token take over (H3, H14, H16, H0) — the name-movers that read and write the indirect object. This is the S-inhibition  $\rightarrow$  name-mover structure of the IOI circuit, spread across roughly eight heads and eight layers rather than concentrated in a few.

This also resolves the apparent null of the per-head tests. The END residual after L22 recovers 59–98% of the prediction, but no single head’s END output does ( $\leq 2\%$ ): the signal arrives via the cumulative S2 $\rightarrow$ END transfer, carried in the residual stream, so ablating or patching any one head leaves the others to reconstruct it.

**Summary.** Contrastive projection, with attention confirmation, correctly identifies the genuine name-mover heads (H14, H16) and their read site (the IO token), and exposes H1 as a sink-head artifact of the norm criterion. Position-resolved patching then traces the full path: an identifiable origin (S2), a transfer window (L18–22), and two head populations (S2-readers then name-movers). The IOI computation in Phi-2 is distributed but fully traceable — no single head is necessary (ablation) or sufficient (per-head patching), because the decision rides the cumulative residual stream. This differs from GPT-2, where [Wang et al. \(2023\)](#) found the circuit concentrated in a few heads; here the same algorithm is spread across many heads and layers, and the contrastive method plus position-resolved patching recovers its structure end to end.

**Accuracy:** 160/160 on Phi-2 (40 name pairs  $\times$  4 templates; mean IO–S logit margin 6.4, always positive), 30/30 on Pythia-1.4B (100%), 29/30 on Pythia-410M (97%).

## 5.2 Factual recall

**Design:** Contrast prompts requiring different factual answers.

“The Eiffel Tower is located in” vs “The Colosseum is located in”:

- L20: “French, France” vs “ancient, Roman, Alexandria”
- L28: “French, Paris” vs “Rome, Roma, Gladiator”

**Per-head decomposition:** At L24, H13 carries the largest contrastive norm (3.9, reads “France, French, Paris”). At L28, the signal distributes across multiple heads (H21, H7, H0), consistent with the factual content spreading from a concentrated source to a broader representation.

**Causal path trace.** To test whether this concentration is causal rather than correlational, we use a single-token contrast — “The capital of **France** is” ( $\rightarrow$  Paris) vs “The capital of **Japan** is” ( $\rightarrow$  Tokyo) — which differ in exactly one input token, the subject. All causal divergence therefore originates at the subject position, so position-resolved patching (restore the clean residual stream at a chosen

position and layer into the corrupt run; measure recovery of  $P(\text{Paris})$  traces where the answer is built and when it moves. CLEAN  $P(\text{Paris}) = 0.806$ , CORRUPT  $= 0.001$ .

| layer  | patch SUBJ only | patch END only | SUBJ+END |
|--------|-----------------|----------------|----------|
| L0–L16 | ~100%           | ~0%            | ~100%    |
| L18    | 99%             | 10%            | 100%     |
| L20    | 74%             | 42%            | 100%     |
| L22    | 50%             | 57%            | 100%     |
| L24    | 8%              | 89%            | 100%     |
| L28    | 0%              | 99%            | 100%     |

**Table 13:** Position-resolved patching for factual recall (recovery of  $P(\text{Paris})$ ). The answer is held at the subject token through the early layers, then handed to END around L22.

The answer lives entirely at the subject position through L16–L18 (a flat plateau: patching the subject alone fully restores Paris), then drains to the END position over L18–L28, with the handoff crossover at **L22**. Patching subject and END jointly recovers ~100% at every layer — the signal rides the cumulative residual along a single subject→END path, with no third site. The plateau-then-move shape matches the subject-MLP enrichment [Meng et al. \(2022\)](#) and [Geva et al. \(2023\)](#) report for factual recall, and the handoff sits a few layers later than IOI’s (L22 vs L18–20).

Position-resolved attention from END names the carriers: attention to the subject peaks at L24, with H16 (0.45) and **H13** (0.31) at the top — and H13 is the same head the contrastive-norm decomposition independently flagged as reading “France/French/Paris.” The norm criterion (“this head’s output differs”) and the causal trace (“this head moves the subject signal to END”) converge on the same unit.

**Generality.** The subject→END structure replicates across 20 country/capital pairs: clean  $P(\text{capital})$  averages 0.88 (all  $\geq 0.5$ ), corrupt  $P \approx 0$ , and the subject→END crossover falls at L22–L24 for every pair (median L24).

**Different factual domains:** “The capital of France is” vs “The capital of Japan is” — the contrastive reads “Paris, France, French” on the France pole and “Japanese, Japan, Tokyo” on the Japan pole. Each entity’s associated knowledge cluster appears on its respective side.

### 5.3 Factual recall vs hallucination

When the model hallucinates — producing a confident but fabricated answer for a fictional entity — the contrastive projection shows what the model’s retrieval surfaces in token space: specific facts for real entities, and nothing beyond name fragments for fictional ones.

**Design:** We construct matched pairs where one prompt elicits genuine recall and the other elicits hallucination, keeping the frame identical (Table 14):

Both sides produce confident, specific answers. But the contrastive projection at L28 reads different content on each pole (Table 15):

| Real prompt                              | Prediction                          | Fictional prompt                                   | Prediction                        |
|------------------------------------------|-------------------------------------|----------------------------------------------------|-----------------------------------|
| Nikola Tesla, born in 1856, invented the | Tesla coil...                       | Ludvig von Vogelkirche, born in 1859, invented the | first practical electric motor... |
| Marie Curie, born in 1867, discovered    | the elements polonium and radium... | Helena Brandström, born in 1871, discovered        | a new species of moth...          |

**Table 14:** Matched real/fictional entity prompts; both elicit confident, specific predictions.

| Pair                 | Real pole (L28)                      | Fictional pole (L28) |
|----------------------|--------------------------------------|----------------------|
| Tesla vs Vogelkirche | Tesla, alternating, Altern, electric | Vog, v, von, Von     |
| Curie vs Brandström  | radio, Radio, Radiation, radioactive | Brand, M, H, a       |

**Table 15:** Contrastive projection at L28: the real pole reads factual associations, the fictional pole only name fragments.

The real pole reads *factual associations* (Tesla → alternating current, Curie → radioactivity). The fictional pole reads *name fragments* (Vog, von, Brand): the readout contains nothing about the entity beyond the tokens of its name and their cultural associations.

**Confirmation via entity-vs-generic contrast.** Contrasting each entity against a bare frame (“A person, born in 1856, invented the”) isolates what the entity name adds. Tesla minus generic reads “Tesla, alternating, AC” at L28 — specific factual content. Vogelkirche minus generic reads “Vog, von, Von” — only name tokens. The readout for the hallucinated entity contains no factual content beyond the name itself.

**Contrastive norm.** The relative norm  $\|\Delta h\| / \|h\|$  at L28 is systematically larger for real-vs-fictional pairs (mean 0.98) than real-vs-real pairs (mean 0.70). Two real entities both occupy a shared subspace of factual-entity representations, constraining their difference; a fictional entity lacks factual content and drifts outside this subspace, producing a larger contrastive norm against any real entity.

**Entropy.** The real inventors predict with lower entropy ( $H = 1.5\text{--}2.5$ ) than their fictional counterparts ( $H = 3.4\text{--}6.3$ ), consistent with the model having specific knowledge to draw on. But entropy alone cannot distinguish “confident recall” from “confident hallucination” — the mountain case illustrates this: Mount Silverhorn (fictional) predicts “2,856 meters” at  $H = 2.8$ , within reach of Mount Everest’s  $H = 1.5$ . The contrastive projection shows the difference: Everest’s pole reads “Nepal, Tibet, Himal” (geographic knowledge), while Silverhorn’s reads “1300, 1100, 1200” (number-range priors calibrated to “Southern Alps,” not specific factual recall).

The contrastive projection does not find a “hallucination flag.” It reads what the model’s retrieval surfaces in token space, and for hallucinated entities that readout contains only name tokens and

contextual priors. A caveat applies to every absence reading: a token missing from the top- $k$  carries less aligned energy than the readout’s visibility threshold, which bounds the signal but is not evidence of zero signal (Tuomi, 2026). What distinguishes the fictional entities is that no factual content surfaces at any layer, where for the matched real entities it does.

## 5.4 Successor heads and temporal structure

All successors are predicted correctly, including wrap-arounds: “After Saturday comes”  $\rightarrow$  Sunday; “After Sunday comes”  $\rightarrow$  Monday; “After December comes”  $\rightarrow$  January.

**Contrastive trajectory:** Contrasting consecutive successors (e.g., “After Monday comes” vs “After Tuesday comes”), the contrastive projection at L24 reads the input day name on its respective pole (“Monday” tokens positive, “Tuesday” negative). By L28 the successor day appears: “Tuesday” on the Monday pole, “Wednesday” on the Tuesday pole. The successor computation is visible as the transition from input-day to output-day content across layers.

**Per-head decomposition at L28:** Decomposing the contrastive signal by attention head identifies H11 as the dominant successor head. Its contrastive norm spikes at the Saturday $\rightarrow$ Sunday and Sunday $\rightarrow$ Monday transitions (Table 16):

| Day pair                              | H11 norm  | Next-largest head |
|---------------------------------------|-----------|-------------------|
| Mon $\rightarrow$ Tue                 | 2         | H20 (3)           |
| Tue $\rightarrow$ Wed                 | 3         | H20 (3)           |
| Wed $\rightarrow$ Thu                 | 3         | H20 (2)           |
| Thu $\rightarrow$ Fri                 | 4         | H20 (5)           |
| Fri $\rightarrow$ Sat                 | 4         | H25 (3)           |
| <b>Sat<math>\rightarrow</math>Sun</b> | <b>11</b> | H12 (4)           |
| <b>Sun<math>\rightarrow</math>Mon</b> | <b>11</b> | H12 (5)           |

**Table 16:** Per-head contrastive norm at L28 for consecutive-day successor contrasts. H11 dominates and spikes at the two week-boundary transitions.

H11’s norm triples at these two transitions; we do not have evidence for what distinguishes them from the weekday transitions. The same head dominates month-pair contrasts at L28, with norms 3–9 across all twelve transitions.

H11 dominates the successor signal but does not carry it alone — H20 and H25 contribute at comparable norms for some day pairs (e.g., H20 norm=5 at Thu $\rightarrow$ Fri vs H11 norm=4).

A separate PCA analysis of the raw hidden states (not using the contrastive projection) finds circular geometry for months and days; this is reported in the supplementary materials as it does not use the paper’s method.

## 5.5 Parametric recall resists counterfactual assertion

When in-context information contradicts parametric knowledge, does the model override its stored facts? We test this by asserting a counterfactual capital and then querying: “The capital of France is

Rome. The capital of France is.”

**The model does not override.** Across four country/capital pairs (France/Rome, Japan/London, Germany/Madrid, Italy/Vienna), the model predicts the correct parametric answer with high confidence despite the counterfactual assertion ( $P(\text{Paris}) = 0.77$  vs 0.81 baseline;  $P(\text{Tokyo}) = 0.87$  vs 0.30 baseline). Even with stronger framing (“It is well established that the capital of France is Rome”), the model predicts Paris at 0.46. Only fictional framing (“In the wizarding world”) partially weakens parametric recall (Paris 0.31, Rome 0.18).

**The contrastive trajectory reads correction, not acceptance.** Contrasting the counterfactual assertion against the veridical one (Table 17):

| Layer | Counterfactual pole reads   | Parametric pole reads |
|-------|-----------------------------|-----------------------|
| L8    | similarly, similar          | [subword fragments]   |
| L20   | Actually, actually, actual  | [subword fragments]   |
| L24   | actually, Actually, correct | centrally, other      |
| L28   | Actually, Greece, Paris     | [comparison tokens]   |

**Table 17:** Contrastive trajectory contrasting a counterfactual capital assertion against the veridical one. From L20 the dominant signal is correction (“Actually”).

From L20 onward, the dominant signal is “Actually, actually, correct” — the model prepares a correction rather than accepting the override. At L28, the Paris token logit is *higher* on the counterfactual side (+9.6) than on the veridical side — the model strengthens its parametric recall when presented with a contradicting assertion. The same holds for all four country pairs.

**Distance weakens correction.** Inserting distractor text between the assertion and query partially weakens the correction reflex: with a medium distractor (~20 words),  $P(\text{Paris})$  drops to 0.38 and  $P(\text{Rome})$  rises to 0.16. The contrastive trajectory at L28 shifts from reading “Actually” to reading the asserted city (“Rome, Italy, Madrid”), suggesting the correction signal decays with distance while the asserted content persists.

**Connection to hallucination (§5.3).** The correction response — “Actually, it is Paris” — requires the model to have retrieved the parametric fact. The hallucination cases in §5.3 show that fictional entities produce no factual associations in the contrastive projection. The correction circuit and the factual-retrieval circuit may share structure: both require that the model has specific knowledge to draw on, and both are visible in the contrastive trajectory as the presence or absence of factual content at mid-to-late layers.

## 6 Contrastive axis taxonomy

Using  $2 \times 2$  factorial designs (crossing two binary axes, e.g., past/future  $\times$  happy/sad), we measure whether each axis produces a consistent contrastive direction across content. Consistency is the cosine between the axis direction extracted from two different content fillers. We test 15 axes across four models (Pythia-410M, Pythia-1.4B, Phi-2, Phi-4).

## 6.1 Three tiers of axis consistency

Axes partition into three tiers:

**Tier 1: Cross-family, token-readable** ( $\cos > 0.7$  in all four models). Eight axes exceed 0.7 consistency in all four models, though some weaken at scale: code/natural language (0.88–0.95), positive/negated (0.84–0.90), past/future (0.81–0.89), English/French (0.83–0.98, declines at scale), assignment/equality-test (0.79–0.92, improves with scale), CAPS/lowercase (0.71–0.87), doubt/certainty (0.71–0.96, weakens at scale), claim/question (0.74–0.83). These axes are near-orthogonal to their content fillers (mean  $|\text{cross-cosine}|$  0.06–0.14).

**Tier 2: Partially consistent** ( $\cos$  0.5–0.8 in some models). Three axes depend on model family or content: formal/informal (strong in Pythia, weak in Phi), thought/speech (moderate, reads as subjective-evaluation vs reporting), cause/effect (weakens at scale).

**Tier 3: Not a direction** ( $\cos < 0.5$  in all models). Four axes never form a consistent direction: animate/inanimate (0.34–0.48), active/passive (0.17–0.65, entangles with content), literal/metaphorical (0.19–0.35, worsens at scale), salient-entity/generic (collapses to  $-0.12$  at Phi-4).

## 6.2 Negation is not cancellation

Projecting different negation types onto the contrastive not direction (Table 18):

| Negation         | Pythia-1.4B | Phi-2      |
|------------------|-------------|------------|
| not              | +43         | +69        |
| never            | +36         | +63        |
| no longer        | +35         | +54        |
| <b>not never</b> | <b>+35</b>  | <b>+66</b> |
| rarely           | +22         | +49        |

**Table 18:** Projection of different negation types onto the contrastive not direction. Double negation (“not never”) does not cancel.

Double negation (“not never”) does not cancel — it projects at 81–95% the strength of single “not.” The model represents “not never” as emphatic negation. Each negation type has its own token readout: not reads as “not, NOT, Not”; no longer reads as “gone, now, replaced” (temporal displacement); rarely reads as “seldom, usually, often” (frequency scale).

## 6.3 Metaphor is processed by domain routing, not a flag

The literal/metaphorical axis has the lowest consistency (0.19–0.35) because metaphor is not a single direction. Instead, each metaphorical use routes to its target domain (Table 19):

**Causal verification (leave-one-out).** For each word, we extract the routing direction from 3 of 4 literal-metaphorical pairs (pairwise  $\cos$  0.64–0.88) and inject into the held-out 4th pair. The held-out readings shift toward the injected domain for all four words (e.g., subtracting the cold literal direction from “The bucket was extremely cold. The temperature was” moves the top predictions

| Contrast                  | Literal pole reads                | Metaphor pole reads              |
|---------------------------|-----------------------------------|----------------------------------|
| cold: ice vs reception    | higher, Celsius, Fahrenheit       | tense, atmosphere, tension, mood |
| sharp: knife vs criticism | blade, blades, stainless          | tone, sarcastic, condescending   |
| bright: lamp vs student   | blinding, overpowering, intensity | proud, gifted, amazed, grades    |
| heavy: boulder vs news    | exceed, load, exert               | mood, tense, gloomy, somber      |

**Table 19:** Literal vs metaphorical contrasts: each metaphorical use routes to its own target domain rather than to a shared figurativity direction.

from “below, so” to “tense, chilly, freezing”), though the shift does not always land on a domain token. The illustrative examples that follow use the direction extracted from all four pairs, so they show causal relevance rather than held-out generalization. Injecting the cold literal direction ( $+1.0\times$ ) into “The reception was extremely cold. The atmosphere was” shifts predictions from “chilly” (social) to “below, freezing” (temperature). Reverse injection ( $-1.0\times$ ) into “The ice was extremely cold. The temperature was” shifts from “below” to “tense” (social). The same pattern holds for sharp, bright, and heavy: each direction bidirectionally flips between the word’s literal and metaphorical domain.

**Dose-response.** The cold domain flip is graded: the metaphorical context shifts through chilly  $\rightarrow$  freezing  $\rightarrow$  below as injection magnitude increases from  $0.25\times$  to  $2.0\times$ . Top-1 prediction flips at  $0.65\times$  of natural magnitude.

**Routing directions are per-domain-pair, not a universal axis.** The cross-domain cosine matrix reveals the structure (Table 20):

|        | cold  | sharp | bright | heavy |
|--------|-------|-------|--------|-------|
| cold   | 1.00  | +0.28 | +0.06  | +0.58 |
| sharp  | +0.28 | 1.00  | +0.03  | +0.37 |
| bright | +0.06 | +0.03 | 1.00   | +0.14 |
| heavy  | +0.58 | +0.37 | +0.14  | 1.00  |

**Table 20:** Cross-domain cosine matrix of the four metaphor routing directions. Cold and heavy share structure; bright is nearly orthogonal to the rest.

Cold and heavy share substantial structure ( $\cos 0.58$ ) — both route between a physical-measurement domain and an emotional domain. Sharp and cold share less (0.28). Bright is nearly orthogonal to all others (0.03–0.14) because it routes to intelligence, a different target domain. The routing direction is determined by the pair of domains being mapped between, not by a shared literalness feature. Cross-domain injection confirms this: injecting the cold routing direction into a sharp-metaphorical context pushes predictions toward temperature tokens (“below, lower”), not toward blade tokens — each direction routes to its respective literal domain.

This explains why literal/metaphorical has Tier 3 consistency: there is no single metaphor axis because there is no single target domain. The cross-domain cosines predict that a probing classifier trained on cold-metaphor would partially transfer to heavy-metaphor (shared target domain) but not to bright-metaphor (different target domain); we did not run this test.

## 7 Discussion

### 7.1 What the method does

Contrastive projection cancels what is shared between two inputs and reads the remainder in token space through  $W_U$ . The method locates where a distinction first appears (per-position reading), which component writes it (attention vs MLP decomposition), and which head carries it (per-head decomposition).

The baseline subtraction (§3.5) can be applied at every token position, producing a position  $\times$  layer map of what the model has computed at each site. Since each position can only attend to earlier ones (causal mask), this traces where intermediate conclusions form before reaching the prediction site.

### 7.2 What the readout means

The contrastive projection reads, in token space, the component that distinguishes the two inputs and is aligned with  $W_U$  token directions. The token labels localize the axis of variation; whether the located component carries causal content is tested separately, by injection (§3.1) and dose-response (§3.2).

When the readout produces uninterpretable tokens, the target signal may still be present below the readout’s detection threshold (Tuomi, 2026). Multi-contrast triangulation (§3.3) recovered token-readable causal content in every compositional case we tested. This does not establish that all model computation is token-readable; it establishes that for the contrasts tested, a token-readable component carrying the causal content could be isolated.

### 7.3 Contrast design

The quality of the readout depends on the contrast. A well-chosen pair that varies one axis produces a clean readout. A poorly chosen pair — or one where the target concept is compositional and mixed with pair-specific content — produces tokens that reflect the mixture, not the target. Multi-contrast averaging addresses this for cases we tested, but we have not established how many baselines are sufficient in general, or whether every model computation yields a token-readable component under this technique. Every projection returns tokens; whether they form a coherent reading or token soup is a qualitative judgement, since the method provides no automatic coherence criterion.

### 7.4 Relationship to circuit analysis

The contrastive projection locates phenomena; circuit analysis explains them. Wang et al. (2023) mapped the IOI circuit in GPT-2 via path patching. Our method identifies heads with the same functional signature in Phi-2 (H14, H1, H16 at L24) by contrastive norm, but does not establish their causal roles. The method is closest to causal tracing (Meng et al., 2022): it identifies where information concentrates, then hands off to heavier tools for causal verification.

## 7.5 Limitations

- **Curated pairs, not sampled.** All demonstrations use hand-constructed minimal pairs. The multi-contrast triangulation uses hand-selected baselines.
- **LayerNorm bypassed.** We skip the final LayerNorm, so  $W_U$  receives vectors at the wrong scale. Token rankings are empirically invariant to this choice, but raw contrastive norms are not comparable across layers due to residual-stream norm growth.
- **$W_U$  readability not guaranteed.** The difference of two states was never trained for  $W_U$  projection. Token labels at intermediate layers are  $W_U$ 's nearest-neighbour assignments. A quantitative criterion for when a component of the difference vector surfaces in this readout is developed in a companion paper (Tuomi, 2026).
- **Smoothness is not  $W_U$ -specific.** Trajectory coherence (consecutive-layer cosine) is a property of  $\Delta h$ , not of  $W_U$ .
- **Exploratory, not causal.** The per-position trace and per-head decomposition identify large contributors, not causes. We verify causality via activation patching for the hot dog case (§4.1), via injection recovery for four cases (§3.1), and via position-resolved patching for IOI and factual recall (§5.1–5.2); per-head claims outside those cases are observational.
- **Coherence is judged by inspection.** Every projection returns tokens; whether they form a coherent reading or token soup is a qualitative judgement. The visibility threshold (Tuomi, 2026) bounds when an aligned component surfaces in the top- $k$ , but provides no automatic coherence criterion.
- **Triangulation coverage.** We tested multi-contrast triangulation on 6 cases. We do not know whether every model computation yields a token-readable component under contrastive subtraction, or how many baselines are sufficient in general.
- **Per-position reading requires tokenization alignment.** The read position must correspond to the same structural role in both inputs.
- **Model coverage.** Primary mechanistic results (§4) on Phi-2 only. IOI replicates on Pythia models. The axis taxonomy (§6) covers four models and shows both consistent and model-dependent patterns.

## 8 Related work

**Contrastive activation methods.** RepE (Zou et al., 2023), ActAdd (Turner et al., 2023), CAA (Rimsky et al., 2024) use matched-pair subtraction for steering. Du et al. (2026) apply contrastive differences projected through the logit lens to trace meta-cognitive control in R1-style reasoning models. Li et al. (2024) use contrastive pairs to study internal representations of truthfulness. Ma et al. (2026) contrast contextualized and non-contextualized logits to locate and rectify conflict-inducing layers;

they intervene to resolve knowledge conflicts, whereas we use the same contrast only to read. We use the same arithmetic for reading, with per-position tracing and attention/MLP decomposition.

**Logit and tuned lens.** [nostalgebraist \(2020\)](#), [Belrose et al. \(2023\)](#). Project individual states through  $W_U$ . [Pal et al. \(2023\)](#) (Future Lens) show a single hidden state also carries information about tokens several positions ahead. The contrastive projection reads the content that differs between two inputs — a different subspace from what the logit lens shows for either input individually.

**Probing and linear representations.** Linear probes ([Belinkov, 2022](#)) train classifiers on hidden states to detect features. Our method achieves a related decomposition without training — the contrastive subtraction is a zero-shot linear probe along the axis defined by the input pair. [Burns et al. \(2023\)](#) extract truth directions from contrast pairs using unsupervised methods; our multi-contrast triangulation is a related technique applied to arbitrary semantic axes rather than truth specifically.

**Superposition and sparse autoencoders.** [Elhage et al. \(2022\)](#) characterized superposition in toy models. [Bricken et al. \(2023\)](#) and [Templeton et al. \(2024\)](#) use sparse autoencoders to decompose superposed representations into monosemantic features. Our multi-contrast triangulation achieves a related isolation — one signal at a time — using paired inputs rather than learned dictionaries. [Lange et al. \(2026\)](#) report that the sparsity objectives used to train cross-layer transcoders can incentivize circuits that match behavior while rewriting deep computation into shallow form. Our reading has no learned dictionary and no sparsity penalty, so it is not subject to that particular incentive; it also does not recover circuit topology, and we make no broader faithfulness claim.

**Circuit analysis and causal tracing.** [Wang et al. \(2023\)](#) reverse-engineered the IOI circuit in GPT-2 via path patching. [Meng et al. \(2022\)](#) used causal tracing to localize factual associations. [Gould et al. \(2024\)](#) identified successor heads via attention pattern analysis. [Conmy et al. \(2023\)](#) automated circuit discovery via ACDC. Our method recovers the key observational findings from these papers (which layers, which heads, which content) but does not establish causality — it is an exploratory complement to these causal techniques.

**Factual recall and knowledge editing.** [Geva et al. \(2023\)](#) traced factual associations to MLP layers. [Hernandez et al. \(2024\)](#) studied linearity of relational knowledge. Our hallucination readings (§5.3) and parametric correction (§5.5) results connect to this line: the contrastive method reads what the retrieval surfaces in token space, and for fictional entities nothing beyond name tokens surfaces.

## Use of AI Assistants

Large language models (Claude, Anthropic; Gemini, Google) were used as assistive tools for coding, running experiments, and drafting text. All research questions, experimental design, and reported claims were directed and verified by the author, who takes full responsibility for the content.

## Code and Data

Code and data are available at <https://github.com/EvidentSolutions/llm-interp/tree/main/contrastive> and archived at <https://doi.org/10.5281/zenodo.20843136>.

## Changes in Version 2

This revision removes interpretive claims that were superseded by subsequent work, without adding new experimental claims. Removed: the “desuperposition” framing (title and throughout); the claim that  $W_U$  token labels name the model’s internal representations; the MLP neuron-correspondence validation (its selection criterion requires a random-initialization control that the experiment lacked); the axis-consistency/frame-forcing table; and the readability-causality head correlation. Absence-based inferences (“the retrieval is empty”) are restated as bounded readout observations. Added: a same-frame near-neighbour causal control, larger replications for IOI (160/160) and factual recall (20 country/capital pairs), and pointers to the companion paper that derives the readout’s visibility threshold (Tuomi, 2026). Two table captions that were swapped in version 1 (successor heads and counterfactual assertion) are fixed.

A second comparative pass against the conference version of this paper re-verified all numbers that differed between the two, by re-running the underlying scripts. Fixed as a result: the LayerNorm-bypass justification and its numbers are replaced with the verified measurement (per-layer mean  $\cos \approx 0.95$ , negation dipping to 0.73 at L31, Figure 1); the top-1 eatability dose table is removed (the logged top-1 at  $+0.5\times$  is “more,” not “tasty”); three greedy continuations are corrected against a fresh deterministic run; the truth-direction experiment is upgraded to the 15-pair leave-one-out replication, with the extraction-set overlap of the illustrative tables now stated; and a factual-recall attention-weight misstatement (“led by H13”) is corrected. Softened or disclosed: the abstract’s random-null z-scores are replaced by the near-neighbour control, the scalar-implicature and metaphor claims are restated at the strength the body supports, and the case-selection process is disclosed. One caveat is withdrawn: the metaphor leave-one-out experiment is genuinely held-out (re-verified by rerun); the extraction-overlap note now applies only to the quoted illustrative examples. Ported from the conference version: the pair-specificity framing, the read-position/preamble caveat, the baseline-subtraction technique, the explanation of above-100% recovery, and the distributed-write neuron/head detail for the compound-noun circuit.

## References

- Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1), 2022.
- Nora Belrose, Igor Ostrovsky, Lev McKinney, Zach Furman, Logan Smith, Danny Halawi, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*, 2023.

Trenton Bricken et al. Towards monosemanticity: Decomposing language models with dictionary learning. Transformer Circuits Thread, Anthropic, 2023.

Collin Burns et al. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations (ICLR)*, 2023.

Arthur Conmy et al. Towards automated circuit discovery for mechanistic interpretability. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Yi Du et al. From latent signals to reflection behavior, 2026.

Nelson Elhage et al. Toy models of superposition. Transformer Circuits Thread, Anthropic, 2022.

Mor Geva et al. Dissecting recall of factual associations in auto-regressive language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

Rhys Gould et al. Successor heads: Recurring, interpretable attention heads in the wild. In *International Conference on Learning Representations (ICLR)*, 2024.

Evan Hernandez et al. Linearity of relation decoding in transformer language models. In *International Conference on Learning Representations (ICLR)*, 2024.

Gerard Lange et al. Cross-layer transcoders are incentivized to learn unfaithful circuits. LessWrong, 2026.

Kenneth Li et al. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Xin Ma et al. CoRect: Context-aware logit contrast for hidden state rectification to resolve knowledge conflicts. *arXiv preprint arXiv:2602.08221*, 2026.

Kevin Meng et al. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

nostalgebraist. Interpreting GPT: The logit lens. LessWrong, 2020.

Koyena Pal et al. Future lens: Anticipating subsequent tokens from a single hidden state. In *Proceedings of the 27th Conference on Computational Natural Language Learning (CoNLL)*, 2023.

Nina Rimskey et al. Steering Llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.

Shangwen Sun, Alfredo Canziani, Yann LeCun, and Jiachen Zhu. The spike, the sparse and the sink: Anatomy of massive activations and attention sinks. *arXiv preprint arXiv:2603.05498*, 2026.

Adly Templeton et al. Scaling monosemanticity: Extracting interpretable features from Claude 3 sonnet. Transformer Circuits Thread, Anthropic, 2024.

Olli Tuomi. A visibility threshold for top- $k$  logit-lens readouts. Zenodo preprint, <https://doi.org/10.5281/zenodo.21461945>, 2026.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023.

Kevin Wang et al. Interpretability in the wild: A circuit for indirect object identification in GPT-2 small. In *International Conference on Learning Representations (ICLR)*, 2023.

Andy Zou et al. Representation engineering: A top-down approach to ai transparency, 2023.